Name: ____________________________

Date: ______________________________

Mr. Rawson • Y8 MYP Physics

1.2 – The Law of Reflection

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can define the **normal** as an imaginary line perpendicular to a surface.
- b) I can state that the angle of incidence equals the angle of reflection.
- c) I can draw and label a correct ray diagram for light reflecting off a plane mirror.
- d) I can explain the difference between regular and diffuse reflection.

Theory

When light hits a smooth surface, such as a **plane mirror**, it bounces back. This process is called **reflection**. To understand exactly how the light behaves, we use a reference line called the **normal**. The normal is an imaginary line drawn at 90 degrees (perpendicular) to the surface of the mirror at the point where the light hits. All angles in reflection are measured from this normal line, not from the surface of the mirror itself.

The behaviour of reflected light follows a strict rule known as the Law of Reflection. This law states that the angle between the incoming light ray and the normal is equal to the angle between the outgoing (reflected) ray and the normal. The incoming ray is called the **incident ray**, and the angle it makes with the normal is the **angle of incidence**. The ray that bounces off is the **reflected ray**, and its angle is the **angle of reflection**. Therefore, if you measure the angles carefully, you will always find:

Angle of incidence = ___________ (four underscores)

Word bank: angle of reflection · incident ray · plane mirror · 90 degrees

There are two types of reflection depending on the surface texture. **Regular reflection** occurs on smooth surfaces like mirrors or still water. Here, parallel rays of light hit the surface and remain parallel after reflecting, which creates a clear image. In contrast, **diffuse reflection** happens on rough or irregular surfaces, such as paper or wood. Even though each individual ray still follows the Law of Reflection at its specific point of contact, the surface bumps cause the rays to scatter in many different directions. This is why we can see non-shiny objects from any angle, but we cannot see our reflection in a piece of paper.

Activity

Investigating the Law of Reflection

You will need a ray box (or laser pointer), a plane mirror, a protractor, and a sheet of white paper.

- Draw a straight horizontal line across the middle of your paper. This represents the surface of the mirror.
- Use your protractor to draw a line perpendicular (at 90°) to the mirror line at its centre. Label this line the **normal**.
- Place the mirror vertically along the horizontal line you drew in step 1. Ensure it is stable.
- Shine the ray box so that a single beam of light hits the point where the normal meets the mirror. This is your incident ray.
- Mark two points on the incident ray and two points on the reflected ray using a pencil. Remove the mirror.

- Draw straight lines connecting these points to complete the diagram.
- Use your protractor to measure the angle between the incident ray and the normal (angle of incidence) and the angle between the reflected ray and the normal (angle of reflection).
- Repeat steps 4–7 two more times, changing the angle of the incoming light each time.

Trial	Angle of Incidence (°)	Angle of Reflection (°)	Are they equal? (Yes/No)
1	____________	____________	____________
2	____________	____________	____________
3	____________	____________	____________

Questions

1. Define the term 'normal' in the context of reflection.

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2. If the angle of incidence is 40°, what is the angle between the reflected ray and the surface of the mirror? (Hint: The normal is at 90° to the surface).

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3. Explain why you can see your face in a bathroom mirror but not on a white wall, using the terms 'regular reflection' and 'diffuse reflection'.

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4. A student measures an angle of incidence of 30° and an angle of reflection of 60°. Identify the error in their measurement or calculation.

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Extension

Find out what happens to light when it hits a **periscope**. How many mirrors are typically used, and at what angles are they placed relative to each other to allow you to see over obstacles?

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